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OMIO: A policy-driven Python library for reproducible microscopy image I/O

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Summary

OMIO (*Open Microscopy Image I/O*) is an open-source Python library that provides a semantic I/O layer for multidimensional microscopy data. It turns heterogeneous microscope outputs into a consistent, analysis-ready representation by enforcing canonical OME-compatible axes (TZCYX), preserving dimensionality, normalizing metadata, and writing deterministic OME-TIFF outputs. OMIO also supports large- and remote-file workflows through Zarr-backed lazy access, reusable disk caches, and memory-aware writing. By standardizing image data at the I/O boundary, OMIO supports FAIR-aligned data sharing and reproducible microscopy analysis, facilitates the development of interoperable community tools, and addresses the growing demand for scalable image-data handling in neuroscience and beyond.

Statement of need

Microscopy image analysis depends on correct interpretation of axes, voxel sizes, time increments, and provenance (Huisman et al., 2021). These quantities are encoded differently across vendor formats and reader libraries, and errors can be silent: arrays may remain numerically plausible while channels, time points, or spatial axes are semantically wrong. OME and OME-TIFF address this at the format level (Goldberg et al., 2005; Linkert et al., 2010), and open bioimage informatics has long emphasized interoperable image data access (Swedlow & Eliceiri, 2009). However, practical Python workflows often still require custom metadata heuristics and format-specific handling before image data can be analyzed reproducibly. OMIO targets microscopy researchers and tool developers who need heterogeneous image inputs to become standardized, OME-compatible, analysis-ready data without reimplementing these decisions in every downstream pipeline.

State of the field

Existing Python tools provide important but different layers of microscopy I/O functionality. `tifffile` (Gohlke, 2025) and `czifile` (Gohlke, 2026) expose format-specific pixel and metadata access, while Bio-Formats (CellProfiler developers, 2019) and `aicsimageio` (Brown et al., 2021) emphasize broad format coverage and framework-level abstraction. OMIO's contribution is complementary rather than duplicative: it builds on available readers where appropriate, but adds a lightweight, pure-Python semantic policy layer that enforces canonical axes, metadata normalization, explicit fallbacks, and deterministic OME-TIFF output. This design choice reflects OMIO's aim to standardize the interface between microscopy file access and reproducible downstream analysis rather than to replace existing reader libraries. Table 1 summarizes this scope across selected Python microscopy I/O packages.

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Table 1. Comparison of selected Python microscopy I/O packages and capabilities.

Capability	tifffile	czifile	aicsimageio	Bio-Formats	OMIO
TIFF / OME-TIFF I/O	yes	no	yes	yes	yes
CZI I/O	no	yes	yes	yes	yes
LSM I/O	yes	no	yes	yes	yes
Thorlabs RAW I/O	no	no	no	no	yes
Canonical axes handling	no	no	no	no	yes
Missing axes handling	no	no	no	no	yes
Voxel-size units	partial	partial	yes	yes	yes
Zarr / lazy workflows	partial	no	partial	partial	yes
OME-TIFF write policy	partial	no	partial	yes	yes
BIDS-like batch processing	no	no	no	partial	yes

Software design

OMIO implements a constrained read-normalize-write pipeline exposed through a compact public API:

- `imread` for input normalization,
- `imwrite` for deterministic OME-TIFF output,
- `imconvert` for one-step conversion,
- `open_in_napari` for visualization, and
- `bids_batch_process` for project-level batch workflows.

Format-specific backends extract pixel data and available metadata, but semantic decisions are centralized: axes are completed and reordered to TZCYX, singleton dimensions are preserved, physical calibration fields are normalized, and provenance is stored in a structured metadata dictionary. The same policy is used for single files, folders, and multi-file OME-TIFF series (Linkert et al., 2010), and extends to folder stacks and BIDS-like project trees (Gorgolewski et al., 2016; Musacchio, Antony, Baijal, Crux, et al., 2026).

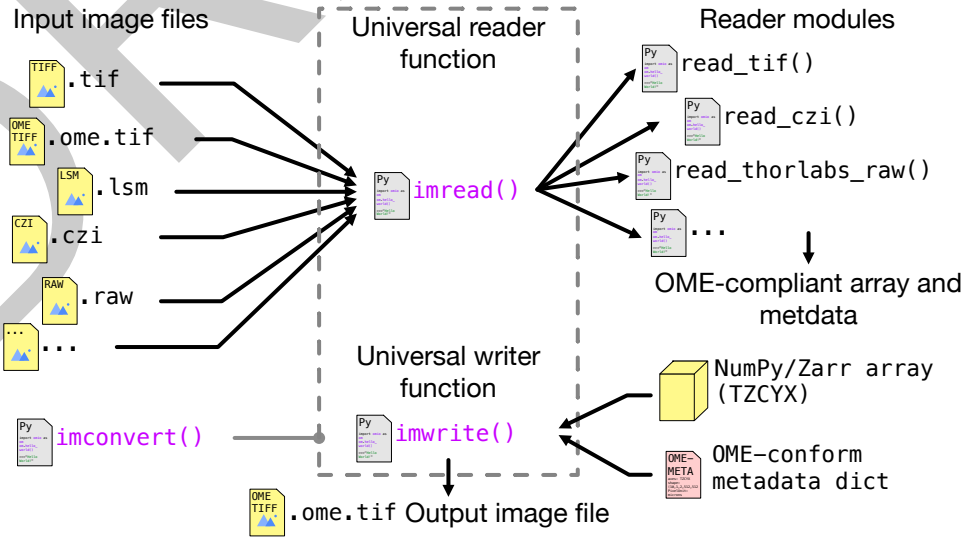


Figure 1: OMIO architecture. Format-specific readers perform low-level extraction, while OMIO enforces axis normalization, metadata standardization, fallback handling, and deterministic OME-TIFF output. Optional Zarr-backed caching supports memory-aware processing of large or remote files.

For large datasets, OMIO can return Zarr arrays (Miles et al., 2025) instead of dense NumPy

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arrays (Harris et al., 2020). Disk-backed caches can be reused after validation, and Zarr-backed inputs can be written plane-wise to OME-TIFF to avoid full-store materialization. Beyond reading and writing, OMIO integrates with Napari (Sofroniew et al., 2025) for interactive image inspection, with optional Dask-backed (Dask Development Team, 2016; Rocklin, 2015) preparation for large or remote files.

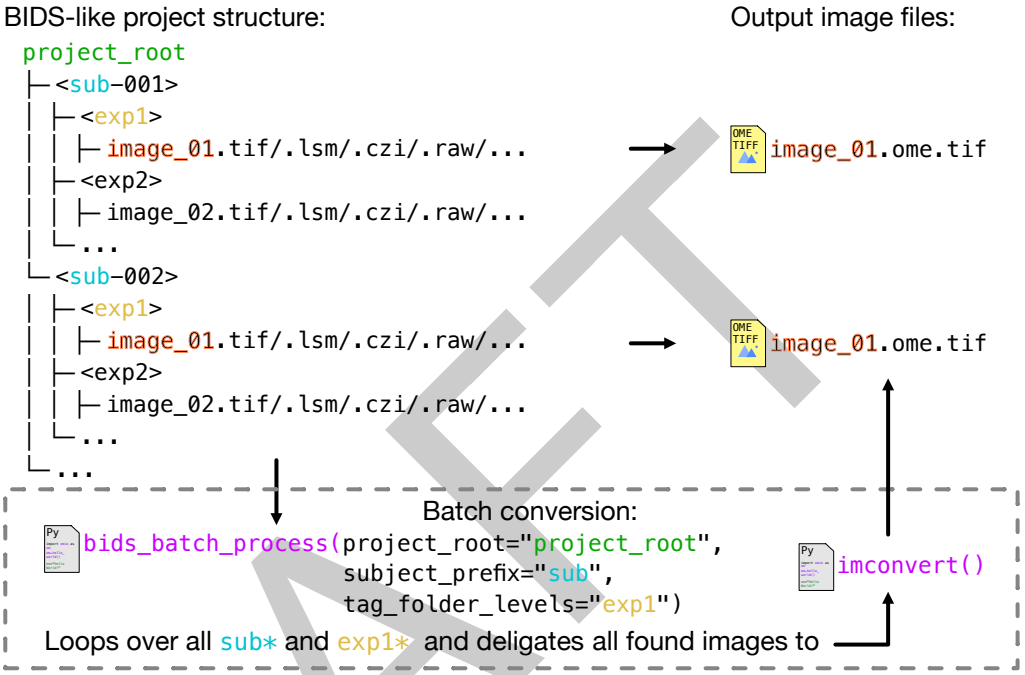


Figure 2: BIDS-like batch processing in OMIO. Microscopy projects organized by subjects, experiments, or acquisition folders can be processed in one batch run, while each discovered image is converted through the same OMIO read-normalize-write pipeline.

Research impact statement

OMIO is intended for reproducible microscopy I/O in Python workflows, especially when heterogeneous inputs must become analysis-ready OME-TIFF with consistent axes and metadata. By making axis semantics, calibration metadata, provenance, software versions, and example data explicit, OMIO supports FAIR-aligned reuse of microscopy data and workflows (Poldrack et al., 2017; Wilkinson et al., 2016). Downstream processing and analysis pipelines can rely on OMIO's standardized axis order and metadata schema instead of guessing dimensions or reimplementing format-specific handling. This provides an auditable foundation for tools that need predictable image semantics across formats, software versions, and project layouts.

Pre-release and current OMIO workflows have supported published and preprint microscopy studies, datasets, and analysis tools (Crux et al., 2024; Fuhrmann et al., 2025; Gockel et al., 2025, 2026; Musacchio et al., 2025; Nebeling et al., 2026). OMIO is also used as an I/O layer in related open-source packages, including ZenReg, spectral-unmixing, and CellColoc (Musacchio, Antony, Baijal, Hoffmann, et al., 2026; Musacchio & Fuhrmann, 2026b, 2026a).

Extensibility

OMIO is modularized into focused reader, writer, cache, viewer, conversion, and batch-processing components. This structure makes it straightforward to add new format backends when pure-Python readers become available or community use cases arise. Contributions are welcome, especially additional readers, representative example files, metadata edge cases, and

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documentation improvements.

Availability

OMIO is released under GPL-3.0. Source code, documentation, tutorials, and issue tracking are available at https://github.com/FabrizioMusacchio/OMIO. The package is distributed via PyPI, documented on Read the Docs (https://omio.readthedocs.io), and archived on Zenodo together with example datasets (Musacchio, 2025b, 2025a).

AI usage disclosure

The authors used OpenAI ChatGPT (GPT-5) to assist with language editing and wording refinement. GitHub Copilot was used for code autocompletion and commit-message assistance. All AI-assisted text and code outputs were reviewed, verified, and edited by the authors, who take full responsibility for the accuracy, integrity, and content of the submitted manuscript and code.

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References

Brown, J. M., Sherman, J., toloudis, Swain-Bowden, M., Chaudhuri, B., Johnson, G., Casero, R., & Hürlimann, D. (2021). *AllenCellModeling/aicsimageio: Mosaics, xarray, and writers, oh my!* (Version v4.0.0). Zenodo. https://doi.org/10.5281/zenodo.4906609

CellProfiler developers. (2019). *Python-bioformats* (Version 1.5.2). https://github.com/CellProfiler/python-bioformats

Crux, S., Roggan, M. D., Poll, S., Nebeling, F. C., Schiweck, J., Mittag, M., Musacchio, F., Steffen, J., Wolff, K. M., Baral, A., Witke, W., Gurniak, C., Bradke, F., & Fuhrmann, M. (2024). Deficiency of actin depolymerizing factors ADF/Cfl1 in microglia decreases motility and impairs memory. *bioRxiv*. https://doi.org/10.1101/2024.09.27.615114

Dask Development Team. (2016). *Dask: Library for dynamic task scheduling*. http://dask.pydata.org

Fuhrmann, F., Nebeling, F. C., Musacchio, F., Mittag, M., Poll, S., Müller, M., Giovannetti, E. A., Maibach, M., Schaffran, B., Burnside, E., Chan, I. C. W., Lagurin, A. S., Reichenbach, N., Kaushalya, S., Fried, H., Linden, S., Petzold, G. C., Tavosanis, G., Bradke, F., & Fuhrmann, M. (2025). Three-photon in vivo imaging of neurons and glia in the medial prefrontal cortex with sub-cellular resolution. *Communications Biology*, *8*(1), 795. https://doi.org/10.1038/s42003-025-08079-8

Gockel, N., Nieves-Rivera, N., Druart, M., Jaako, K., Fuhrmann, F., Musacchio, F., Antony, H., Mittag, M., Crux-Daseking, S., Poll, S., Jansone, B., Fuhrmann, M., & Le Magueresse, C. (2026). Schizophrenia-associated complement C4 impairs synaptic connectivity and decreases microglia-synapse interactions through CR3 signaling. *Cell Reports*, *45*(4), 117161. https://doi.org/10.1016/j.celrep.2026.117161

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Gockel, N., Nieves-Rivera, N., Druart, M., Jaako, K., Fuhrmann, F., Rožkalne, R., Musacchio, F., Poll, S., Jansone, B., Fuhrmann, M., & Magueresse, C. L. (2025). *Example datasets for microglial motility analysis using the MotilA pipeline*. Zenodo. https://doi.org/10.5281/zenodo.15061566

Gohlke, C. (2025). *Cgohlke/tifffile: v2025.12.20* (Version v2025.12.20). Zenodo. https://doi.org/10.5281/zenodo.18005696

Gohlke, C. (2026). *Cgohlke/czifile* (Version v2026.8.16). Zenodo. https://doi.org/10.5281/zenodo.21953612

Goldberg, I. G., Allan, C., Burel, J.-M., Creager, D., Falconi, A., Hochheiser, H., Johnston, J., Mellen, J., Sorger, P. K., & Swedlow, J. R. (2005). The open microscopy environment (OME) data model and XML file: Open tools for informatics and quantitative analysis in biological imaging. *Genome Biology*, *6*(5), R47. https://doi.org/10.1186/gb-2005-6-5-r47

Gorgolewski, K. J., Auer, T., Calhoun, V. D., Craddock, R. C., Das, S., Duff, E. P., Flandin, G., Ghosh, S. S., Glatard, T., Halchenko, Y. O., Handwerker, D. A., Hanke, M., Keator, D., Li, X., Michael, Z., Maumet, C., Nichols, B. N., Nichols, T. E., Pellman, J., … Poldrack, R. A. (2016). The brain imaging data structure, a format for organizing and describing outputs of neuroimaging experiments. *Scientific Data*, *3*(1), 160044. https://doi.org/10.1038/sdata.2016.44

Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk, M. H. van, Brett, M., Haldane, A., Río, J. F. del, Wiebe, M., Peterson, P., … Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*(7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

Huisman, M., Hammer, M., Rigano, A., Boehm, U., Chambers, J. J., Gaudreault, N., North, A. J., Pimentel, J. A., Sudar, D., Bajcsy, P., Brown, C. M., Corbett, A. D., Faklaris, O., Lacoste, J., Laude, A., Nelson, G., Nitschke, R., Grunwald, D., & Strambio-De-Castillia, C. (2021). *A perspective on microscopy metadata: Data provenance and quality control*. https://doi.org/10.48550/arXiv.1910.11370

Linkert, M., Rueden, C. T., Allan, C., Burel, J.-M., Moore, W., Patterson, A., Loranger, B., Moore, J., Neves, C., MacDonald, D., Tarkowska, A., Sticco, C., Hill, E., Rossner, M., Eliceiri, K. W., & Swedlow, J. R. (2010). Metadata matters: Access to image data in the real world. *Journal of Cell Biology*, *189*(5), 777–782. https://doi.org/10.1083/jcb.201004104

Miles, A., Kirkham, J., Stansby, D., Papadopoulos Orfanos, D., Hamman, J., Bennett, D., Bussonnier, M., Moore, J., Augspurger, T., Cherian, D., Jones, M., Rzepka, N., Spitz, H., Verma, S., Bourbeau, J., Fulton, A., Abernathey, R., Lee, G., Kristensen, M. R. B., & Durant, M. (2025). *zarr-developers/zarr-python: v3.1.5* (Version 3.1.5). Zenodo. https://doi.org/10.5281/zenodo.17672242

Musacchio, F. (2025a). *OMIO Example Datasets: Dummy Microscopy Files Plus Curated Third-Party Samples for I/O Testing* [Data set]. Zenodo. https://doi.org/10.5281/zenodo.18078231

Musacchio, F. (2025b). *OMIO: A policy-driven python library for reproducible microscopy image i/o* (Version v0.3.0). https://doi.org/10.5281/zenodo.22029020

Musacchio, F., Antony, H., Baijal, A., Crux, S., Fuhrmann, F., Gockel, N., Hoffmann, D. M., Mercan, D., Nebeling, F. C., Wolff, K., & Fuhrmann, M. (2026). *OMIO: A policy-driven python library for reproducible microscopy image i/o*. Cold Spring Harbor Laboratory. https://doi.org/10.64898/2026.06.09.731118

Musacchio, F., Antony, H., Baijal, A., Hoffmann, D. M., Nebeling, F. C., Crux, S., & Fuhrmann, M. (2026). *CellColoc: A modular, open-source workflow for cell colocalization, segmentation, and feature extraction in microscopy images*. Cold Spring Harbor Laboratory.

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https://doi.org/10.64898/2026.07.26.740771

Musacchio, F., Crux, S., Nebeling, F., Gockel, N., Fuhrmann, F., & Fuhrmann, M. (2025). MotilA – a python pipeline for the analysis of microglial fine process motility in 3D time-lapse multiphoton microscopy data. *Journal of Open Source Software*, *10*(116), 9267. https://doi.org/10.21105/joss.09267

Musacchio, F., & Fuhrmann, M. (2026a). *Spectral unmixing: A modular and reproducible python package for directed and blind spectral unmixing in multidimensional microscopy stacks*. Cold Spring Harbor Laboratory. https://doi.org/10.64898/2026.07.06.736825

Musacchio, F., & Fuhrmann, M. (2026b). *ZenReg: A modular python platform for fast and memory-efficient n-dimensional microscopy image registration*. Cold Spring Harbor Laboratory. https://doi.org/10.64898/2026.08.07.743572

Nebeling, F. C., Fuhrmann, F., Mittag, M., Musacchio, F., Antony, H., Gockel, N., Friker, L. L., Leonardelli, S., Filser, S., A., D., Stork, M., Bano, D., Pietsch, T., Giordano, F. A., Zhou, Q., Parrinello, S., Hölzel, M., Herrlinger, U., Salomoni, P., & Fuhrmann, M. (2026). Microglia-glioblastoma crosstalk mediates glioblastoma invasion at the far infiltration zone [Doi: 10.1016/j.immuni.2026.03.010]. *Immunity*, *59*(4), 1075–1091.e4. https://doi.org/10.1016/j.immuni.2026.03.010

Poldrack, R. A., Baker, C. I., Durnez, J., Gorgolewski, K. J., Matthews, P. M., Munafò, M. R., Nichols, T. E., Poline, J.-B., Vul, E., & Yarkoni, T. (2017). Scanning the horizon: Towards transparent and reproducible neuroimaging research. *Nature Reviews Neuroscience*, *18*(2), 115–126. https://doi.org/10.1038/nrn.2016.167

Rocklin, M. (2015). Dask: Parallel computation with blocked algorithms and task scheduling. In K. Huff & J. Bergstra (Eds.), *Proceedings of the 14th python in science conference* (pp. 130–136). https://doi.org/10.25080/majora-7b98e3ed-013

Sofroniew, N., Lambert, T., Bokota, G., Nunez-Iglesias, J., Sobolewski, P., Sweet, A., Gaifas, L., Evans, K., Burt, A., Doncila Pop, D., Yamauchi, K., Weber Mendonça, M., Rodríguez-Guerra, J., Liu, L., Buckley, G., Vierdag, W.-M., Anderson, A., Monko, T., Willing, C., & Zhao, R. (2025). *Napari: A multi-dimensional image viewer for python* (Version v0.7.0a0). https://doi.org/10.5281/zenodo.18089604

Swedlow, J. R., & Eliceiri, K. W. (2009). Open source bioimage informatics for cell biology. *Trends in Cell Biology*, *19*(11), 656–660. https://doi.org/10.1016/j.tcb.2009.08.007

Wilkinson, M. D., Dumontier, M., Aalbersberg, Ij. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., da Silva Santos, L. B., Bourne, P. E., Bouwman, J., Brookes, A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R., … Mons, B. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, *3*(1), 160018. https://doi.org/10.1038/sdata.2016.18

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